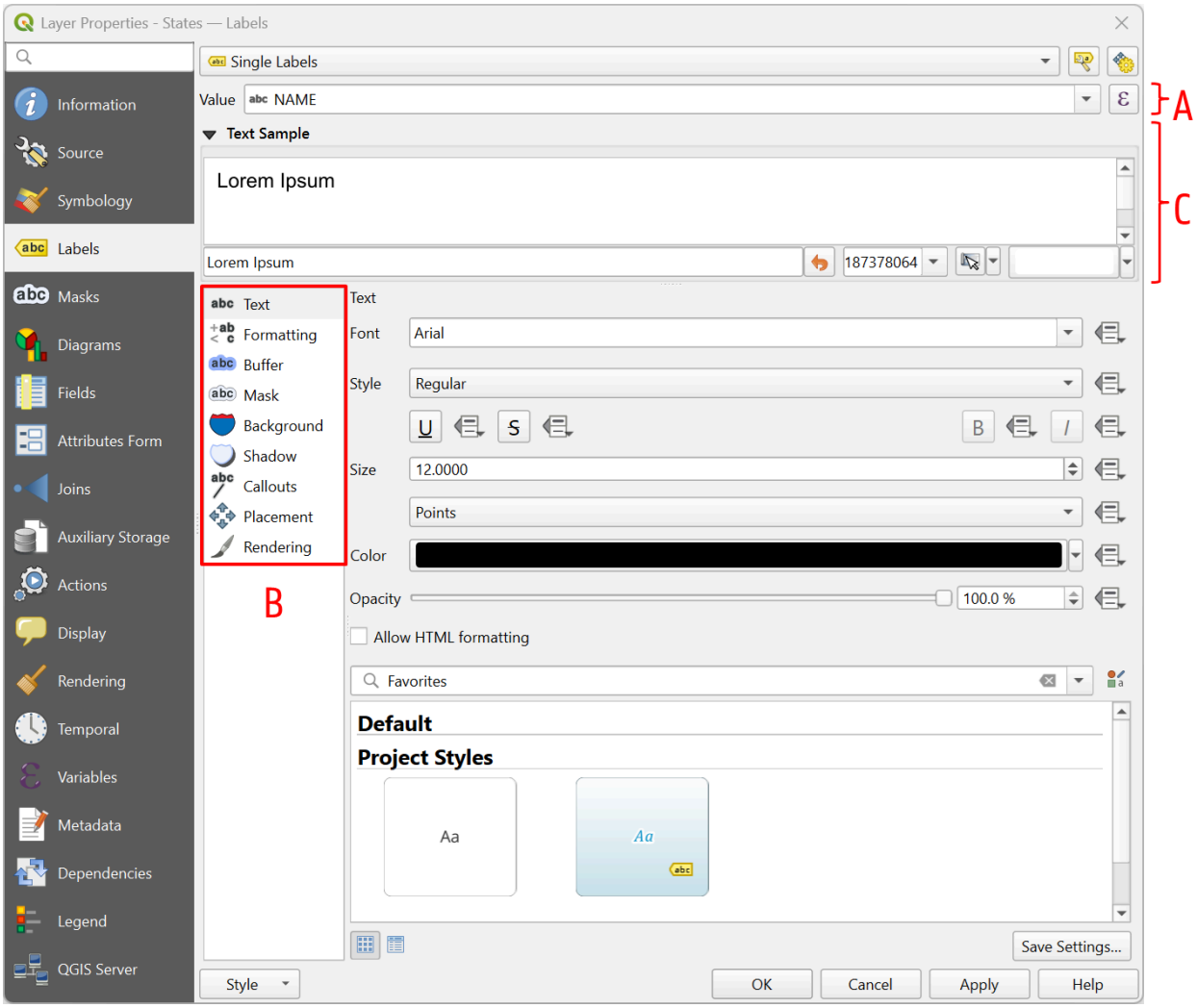
# 4.2.7 Text labels

## Adding text labels automatically

This section covers labels that QGIS adds automatically to vector features, based on the values in their attribute fields.

> **See also**
>
> Adding text to a map by hand is explained later, in the print layouts: see General: layouts.

You set up automatic text labels in the *Labels* tab of your *Layer Properties* window. Choose *Single Labels* in the menu to show the settings below. This default setting gives all features the same label style (font, size...). How to vary the labels between features is explained later, in Rule-based styling.




First, you decide **what** to **label** the features with: choose an attribute field under *Value* (A).

Then you configure the **visual settings**. As with diagrams, they are spread over several tabs (B).

- In the tabs ***Text***, ***Formatting***, ***Buffer***, ***Background*** and ***Shadow***, you set how the text looks: font, line spacing, background... A preview at the top (C) shows the result of the current settings.

- Under ***Callouts*** you choose whether QGIS draws connector lines between the features and their labels. These connectors help when you place the labels further away from the features. The QGIS documentation page gives more information.
- Under ***Placement*** you set where the label is placed relative to its feature. As with diagrams, the options depend on the layer type.

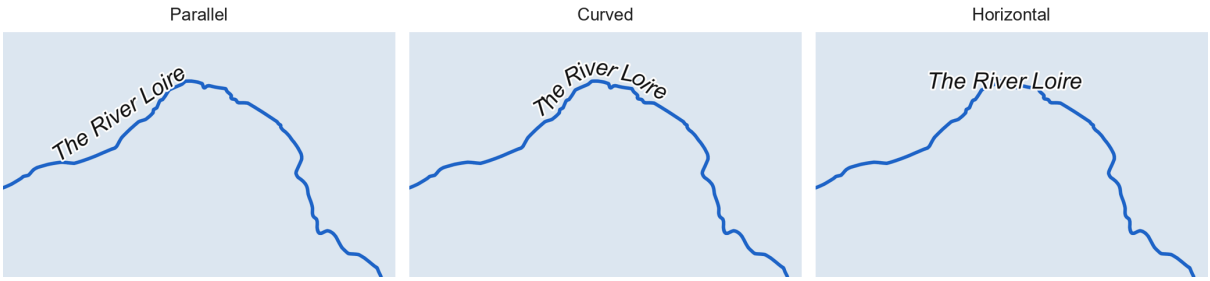
  ▸ **Points**

  With *Cartographic*, the default for points, QGIS tries a set of positions around the point in an order that follows cartographic conventions: first top right, then top left, bottom right, bottom left, right, left… It chooses the first position that does not clash with other labels. As with *Around point*, you set the distance between the point and the label yourself (*Distance*).



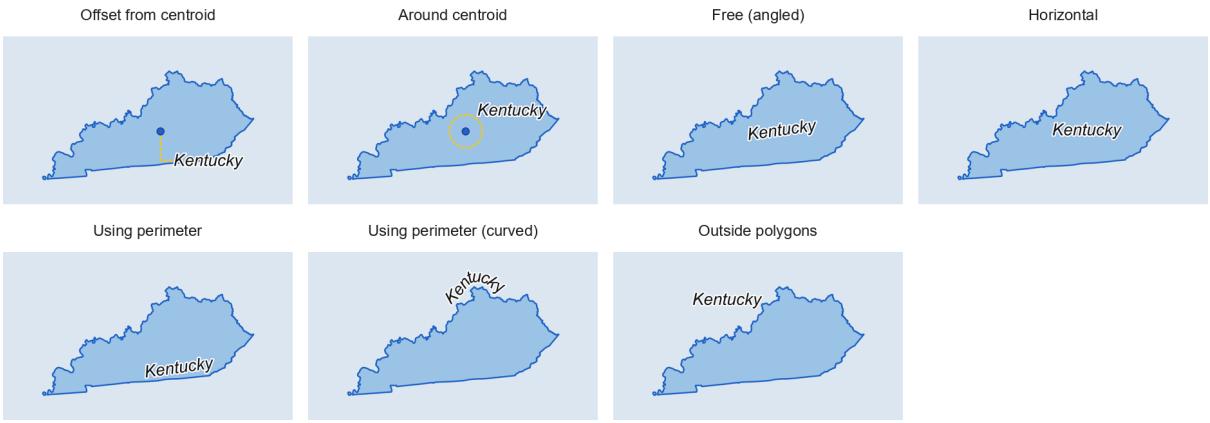

*The label placement options for points in QGIS, shown on Ghent.*

  ▸ **Lines**




*The label placement options for lines in QGIS, shown on the bend of the Loire at Orléans. River: Natural Earth (smoothed).*

  ▸ **Polygons**




*The label placement options for polygons in QGIS, shown on the state of Kentucky; blue dot = centroid.*

There is often not enough room on the map for all labels without overlaps, for example when you zoom out far, or when many small features lie close together. QGIS therefore looks at the labels of all layers together, and **tries to fit as many labels as possible on the map**. Labels with a higher *Priority* (see below) are left out less easily than others. With most placement options (for example *Cartographic*, *Around point* or *Free*), QGIS has some freedom. In that case, QGIS solves a kind of puzzle: it tries several positions for each label so that as many labels as possible fit on the map, and takes into account the preferred position of each label (for example, top right with *Cartographic*). With other options, such as *Offset from point*, you set one fixed position for the label yourself. QGIS then has less freedom to place the labels, so that more labels can end up missing.

Further down the *Placement* tab, a few sections help when labels lie close together:

- *Label Spacing*: a margin around each label in which no other label can be placed.
- *Duplicate Labels*: with *Avoid duplicate labels*, QGIS does not place labels with exactly the same text close to each other, even if they belong to different features or layers. Under *Settings...* you set how close is too close (by default 20 mm on the map). Note that this can also hide labels you do want to see: the river and the state of Mississippi have the same name, so one of the two labels can disappear if they lie close together. Labels on the parts of one multi-part feature are a separate case, see *Feature options* under *Rendering* below.
- *Priority*: where labels compete for the same spot, QGIS shows the one with the higher priority and can leave the others out.
- *Obstacles*: with *Features act as obstacles* (on by default), QGIS tries to avoid placing labels or diagrams on top of the features of this layer. This is a preference, not a strict rule. For polygons, QGIS by default only avoids the boundary, so that labels lie completely inside or completely outside a polygon. Under *Settings...* you can choose to avoid the interior of the polygons instead.

- Finally, under ***Rendering*** you set a few general but important display options.

  Under ***Label options*** you can **limit the scales** at which labels are visible (*Scale dependent visibility*). For example, you can show the names of the major cities only when you have zoomed in far enough, so that they do not crowd the map when you look at the whole country.

  Under ***Overlapping labels*** you choose a *Mode*: *Never Overlap* (the default: a label that would overlap another one is left out), *Allow Overlaps if Required* (QGIS first tries a less ideal position, and only lets the label overlap if there is none) or *Allow Overlaps without Penalty* (QGIS always uses the best position, even if labels overlap).

  > Tip
  >
  > If some labels do not appear on the map, a common reason is that they would overlap other labels: by default (*Never Overlap*) QGIS leaves those out. Zoom in, or change the *Overlapping labels* setting. To see which labels QGIS left out, click *Show Unplaced Labels* in the *Label Toolbar*.

  Under ***Feature options*** you set, among other things, whether labels are only shown when they fit completely inside the feature (polygon layers only), whether there is a maximum number of labels you want to display, and how multi-part features are labelled (*Multipart geometry labeling*). A multi-part feature is a single feature that consists of several separate parts, such as a state with islands.

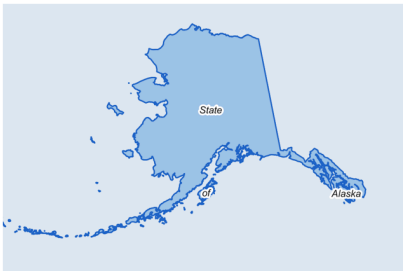
Label Largest Part Only



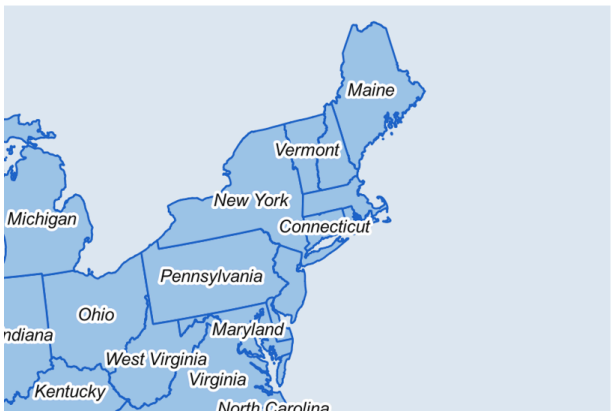

Label Every Part with Entire Label




Split Label Text Lines over Parts




Only draw labels which fit completely within feature: off




… on

